How webservers work: - (Task 3)

What is a web server?

A web server is a software that listens for incoming connections and then utilise HTTP protocol to deliver web content to its clients. The most common web server software you'll come across is Apache, Nginx, IIS and NodeJS. A webserver delivers files from whats called it root directory, which is defined in software settings. For Example Nginx and Apache share same default location of /var/www/html in Linux operating system and IIS uses C:\inetpub\wwwroot for window operating systems. Ex If you requested file http://www.example.com/Picture.jpg, It would send the file /var/www/html/Picture.jpg from its local hard drive.

Virtual Hosts :-

webservers can host multiple websites with different domain names, to achieve this they use virtual hosts. The web server software checks hostname being requested from the HTTP header and matches that against its virtual hosts (virtual hosts are text based configuration files). If its finds a match, the correct website will be provided. If no match the default website will be provided instead.

* Virtual Hosts can have their root directory mapped to different locations on hard drive.
  Ex:- one.com being mapped to /var/www/website_one and two.com is being mapped to /var/www/website_two

* There is no limit to number of different websites you can host on web server.

Static v Dynamic content :-

Static content as name suggests, is content that never changes. Common Example of this picture are js, CSS, etc but can also include HTML that never changes. Furthermore, these are files that are directly served from webserver with no changes made to them.

Dynamic content, on the other hand is content that could change with different requests. Ex:- blog. on homepage of blog. It will show you latest entries. If new entry is created the home page is then updated in latest entry or secondary example might be search page on blog. Depending on what word you search, different results will be displayed. These changes to what you end up seeing are done in what is called the Backend with use of programming and scripting language. It's called backend because what is being done is all done behind the scenes. you can't view websites HTML source and see what's happening in Backend, while HTML is result of processing from Backend. Everything you see in your browser is called frontend.

Scripting and Backend languages :-

There's not much of limit to what a backend language can achieve, and these are what make a website Interactive to user. Ex:- languages are PHP, Python, Ruby, NodeJS, perl and many more. These languages can interact with databases called external services, process data from user and so much more. A very basic PHP example of this would be if you requested website http://example.com/index.php?name=adam

If index.PHP built like this:

```
<html><body> Hello <?PHP echo $_GET["name"]; ?> </body></html>
```

Output of following to client is:

```
<html> <body> Hello adam </body></html>
```

You'll notice that client doesnot see any PHP Code because Its on Backend. This Interactivity opens up lot more security Issues for web applications that haven't been created Securely, as you learn in further modules.

Quiz:-

1) What does web server software use to host multiple sites?

A) Virtual Hosts

2) What is the name for type of content that can change?

A) Dynamic

3) Does the client see backend code? Yay | Nay

A) Nay

Whole module Quiz: –

Request tryhackme.com in your browser → Check local cache for IP Address → Check your recursive DNS server for Address →

Query root server to find authoritative DNS Server → Authoritative DNS Server advises the IP address for website → Request passes through a load Balancer →

Connect to webserver on port 80 or 443 → Web server receives the GET request → Web Application talk to Database → Your Browser renders the HTML into a viewable website